01 / POSTER MAIN FIGURE

Choose histories with enough discount variation

FreshRetailNet links daily sales, discounts and stockouts for each store-product pair.

PUBLIC DATA

50,000

store-product histories

→

PASS THE FILTERS

16,684

eligible histories

→

WORKING COHORT

312

histories in five stores

Require repeated price changes

15-85% discounted days; at least six discount-state switches;
zero sales on fewer than 20% of days.

Select the top five stores

Rank stores by eligible histories.
All five happen to be in one city.

312 histories | 122 products | 25,896 training/validation rows | 2,184 final-week cases

This cohort supports comparison; it does not represent every store or product.

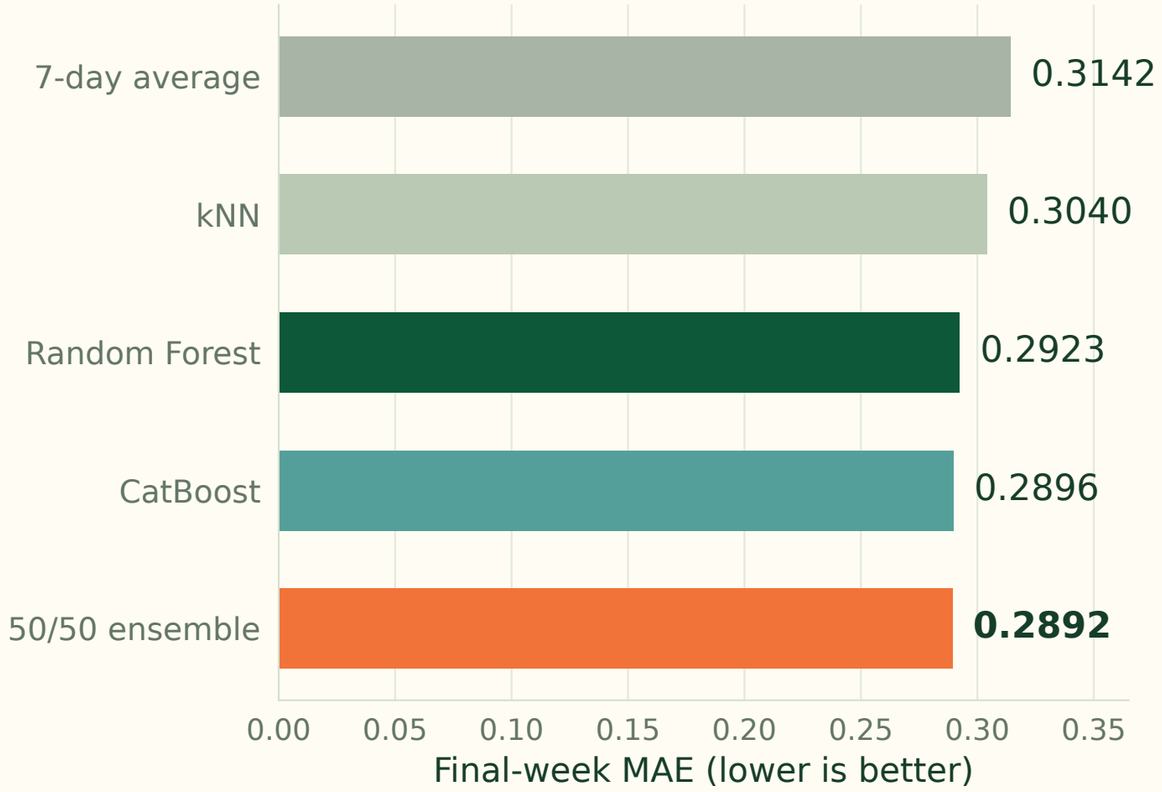
Discounted day: price rate <0.95. Retrospective selection across the 90-day training file. Diagram stages are not area-scaled.

Final notebook / a7e26dd

Forecasts improved; deep-discount days remained difficult

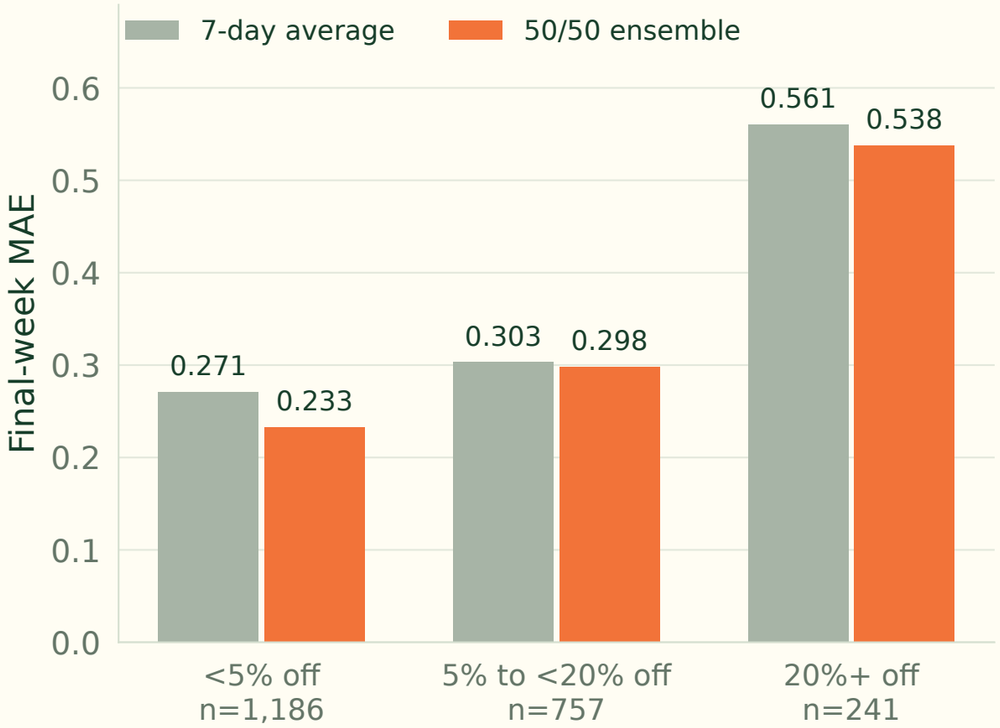
Same 2,184 final-week cases. MAE is mean absolute error in normalized observed sales.

Overall model comparison



7.9% lower MAE than baseline

Errors by observed discount



Higher absolute errors also accompany higher sales.

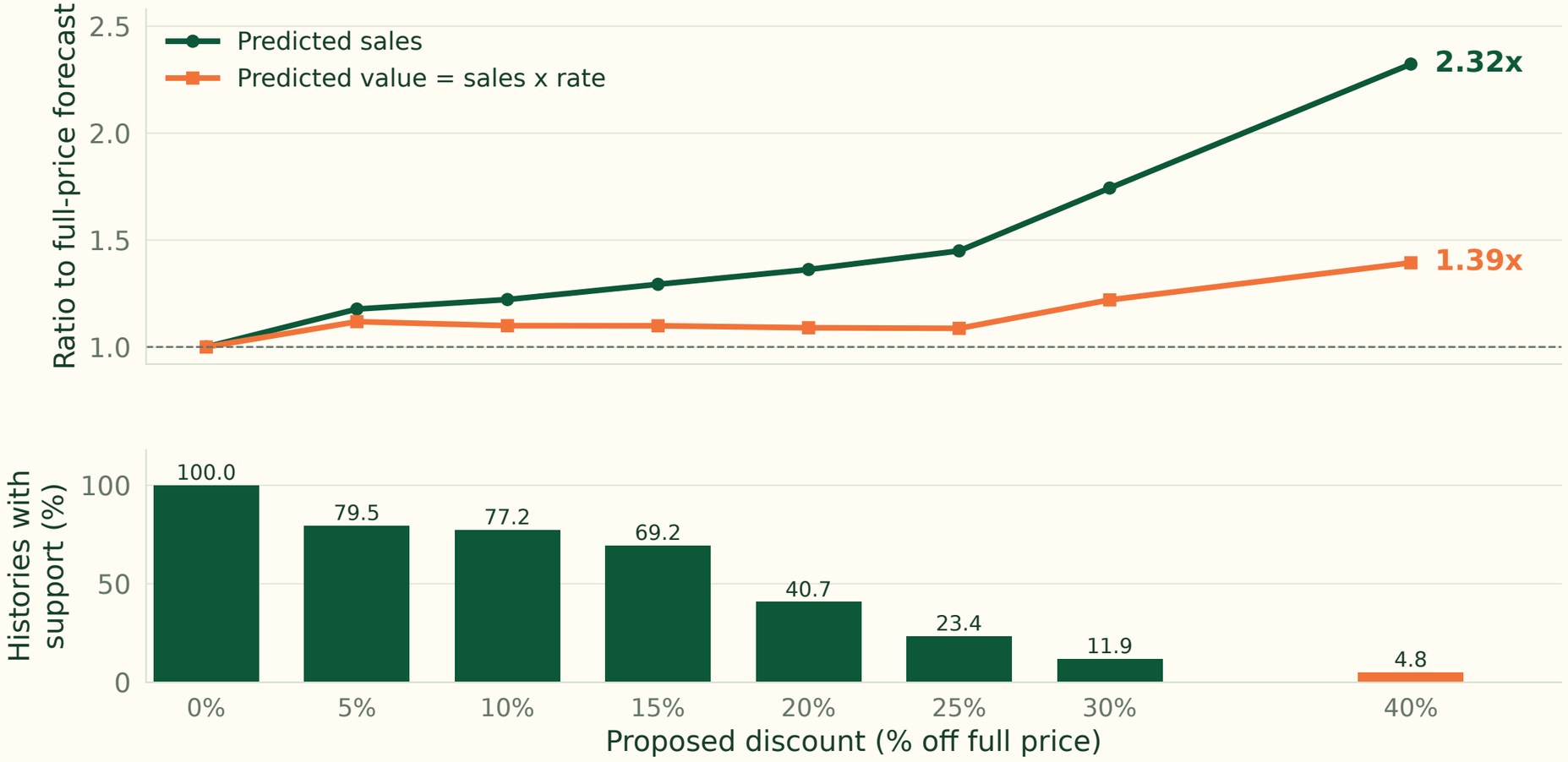
50% Random Forest + 50% CatBoost; CatBoost alone is almost tied.

Final period: 26 Jun-2 Jul 2024, reused during development. Deep-discount MAE is not a scale-free reliability or causal measure.

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The largest predicted gains have the weakest support

Hold each case's history fixed and vary the proposed discount. These are model scenarios.



At 40% off

2.32x

predicted sales

15 / 312

histories support this rate (4.8%)

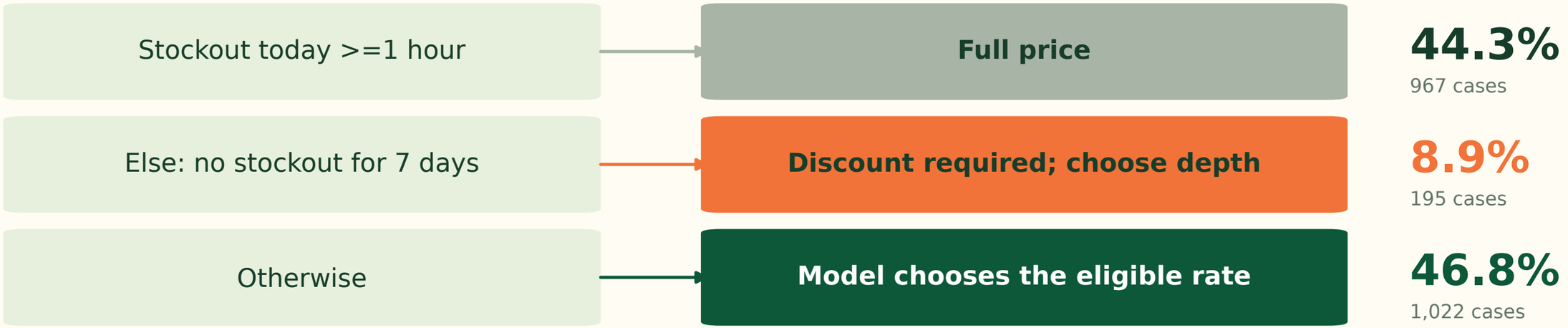
Support requires 3+ training days within 0.025 of the price rate.

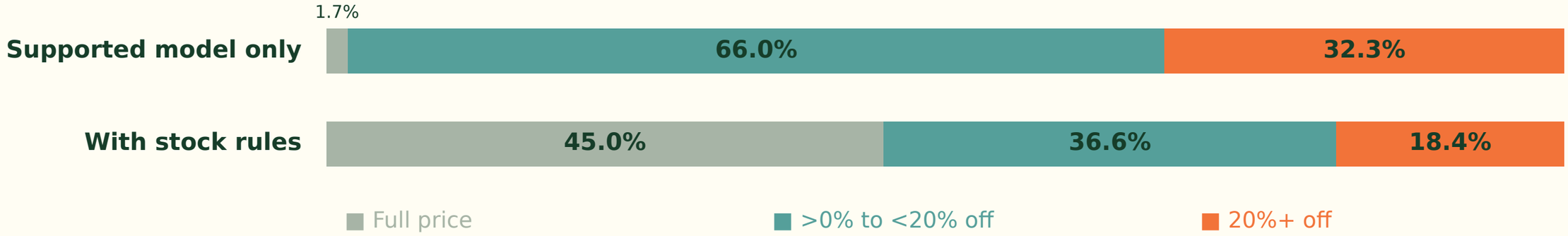
A strong predicted response is not evidence that a new discount will cause the same gain.

Ratios compare means over the same 2,184 cases; support denominator is 312 histories. Zero unit cost. Value is a normalized proxy.

Stock rules change both whether and how much to discount

Restrict to supported rates, apply the assumed rules, then compare eligible predicted values.





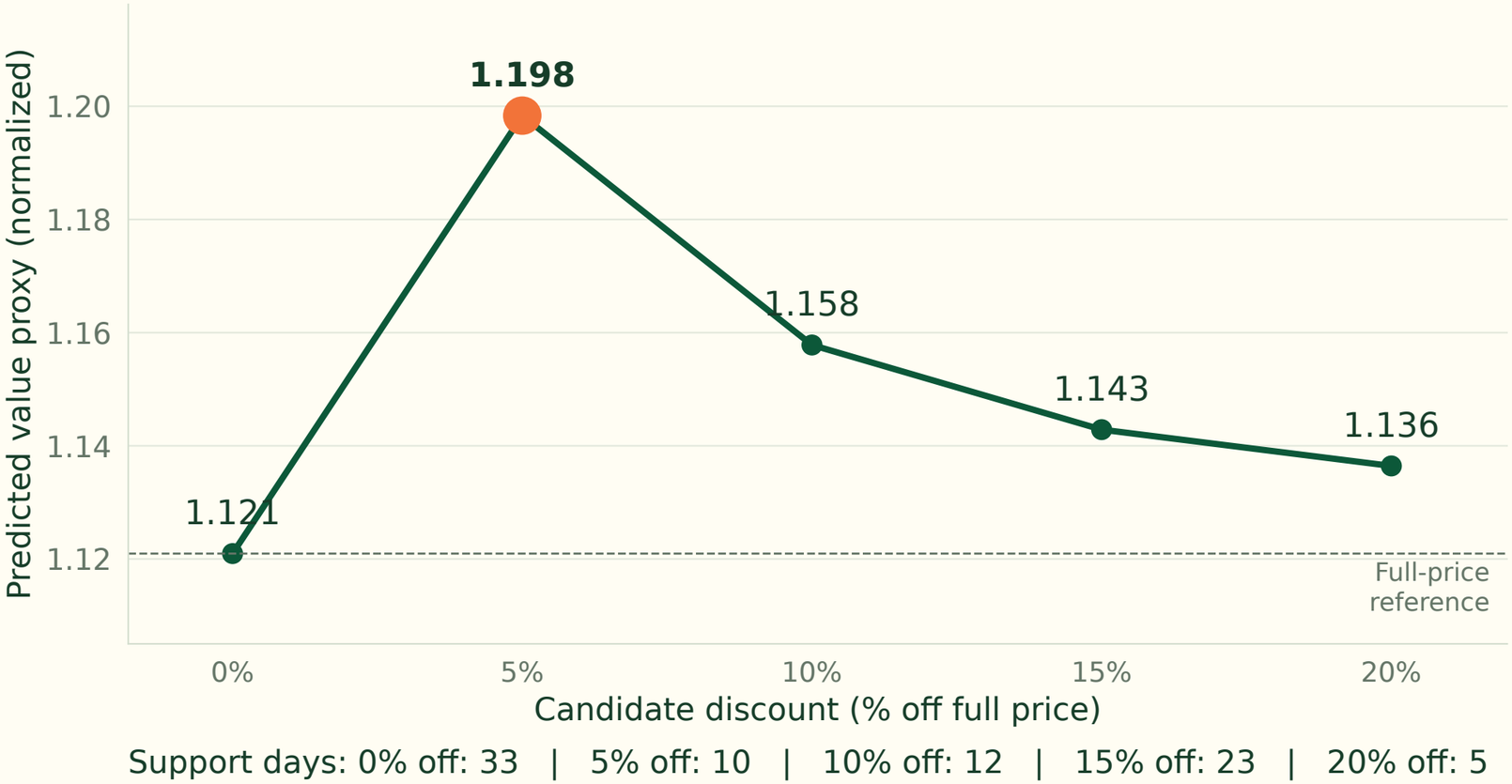
Any discount: 98.3% of cases becomes 55.0%. This is changed behavior, not proven business gain.

Same 2,184 cases, zero assumed unit cost. Stockout history does not establish inventory age or freshness. Notebook Table 5 supplies counts.

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For one product-day, predicted value peaks at 5% off

Store 18 / Product 11 / 26 Jun 2024. Five candidates pass the historical-support check.



Support days: 0% off: 33 | 5% off: 10 | 10% off: 12 | 15% off: 23 | 20% off: 5

5% off

Selected among eligible rates

Price rate 0.95
x predicted sales 1.261
= predicted value 1.198

25%, 30% and 40% off are excluded: no training days within the support window.

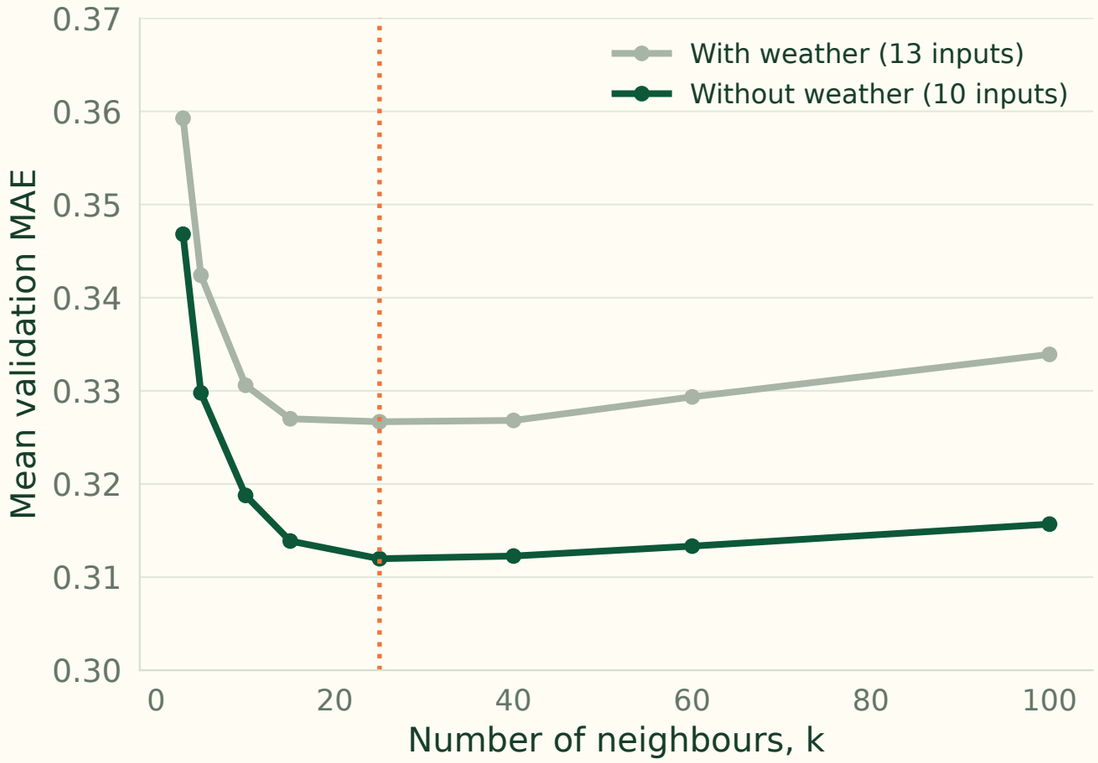
The model supplies a concrete rate; its predicted gain still needs a real-world test.

Zero unit cost. Deterministic first eligible worked case, not selected for a large gain. Proxy values are not currency or realized revenue.

More features and more trees did not always help

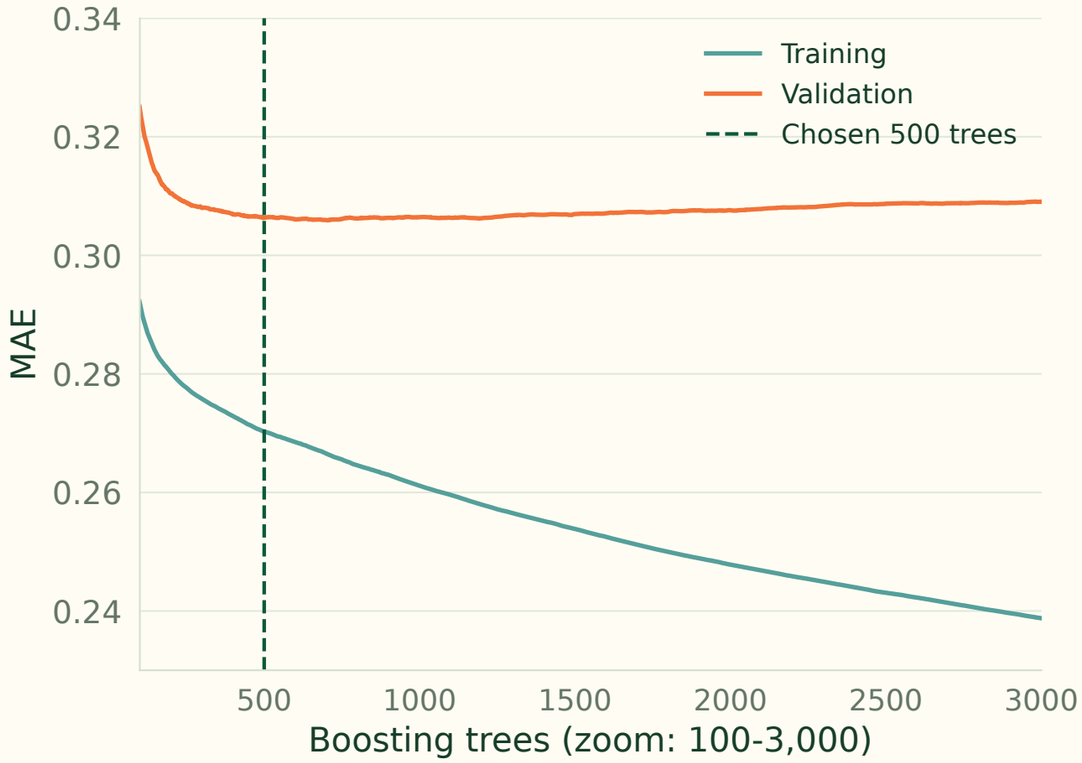
Use this in the notebook or as a supporting figure when explaining the modelling choices.

Removing weather improved kNN



At k=25: 0.3267 -> 0.3120

Training improves; validation levels off



Learning rate 0.03; final validation fold

Choose complexity using later dates, not how well the model fits its training data.

Left: same five time-ordered folds, equal feature weights. Right: recorded CatBoost learning curve; final settings selected across five folds.

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